Blinkit delivery reliability: objective and measurement

Milestone 3 • Data, Metrics and Outcomes Mapping • 1,500 orders, 20 stores, 5 cities, 1–14 Sep 2026

OBJECTIVE

To keep delivery promises realistic and reduce avoidable delays and company-caused cancellations to deliver more orders on time.

NORTH STAR **On time fulfilment rate**

Orders delivered by the original deadline ÷ eligible orders × 100

A ÷ E × 100, where E = A + B + C

Rule	What it means
A / B / C	A / B / C: A = on time; B = late; C = company-caused cancellations: no rider (18) or unavailable items (8). Eligible orders: E = A + B + C = 1,481.
The clock	From order placement to customer handover, we judged against the original checkout promise, but it is not a revised ETA. Exactly on the deadline counts as on time. Timestamp meanings need confirmation.
Not yet counted	19 cancellations—payment issues (8), customer cancelled (11). Their underlying causes remain unclear.
Data limits	Stage times are estimates. How these 1,500 orders were selected from Demand is not recorded, so findings describe these orders only.

To verify: What caused the 19 cancellations, the timestamps show customer handover and the original checkout promise, and how the 1,500 orders were selected from Demand's 128,528 orders

PROVISIONAL NORTH STAR

73.7% on-time fulfilment (1,092 ÷ 1,481)

Late: 363 orders (24.5% of E)

On time among delivered: 75.1% (1,092 ÷ 1,455)

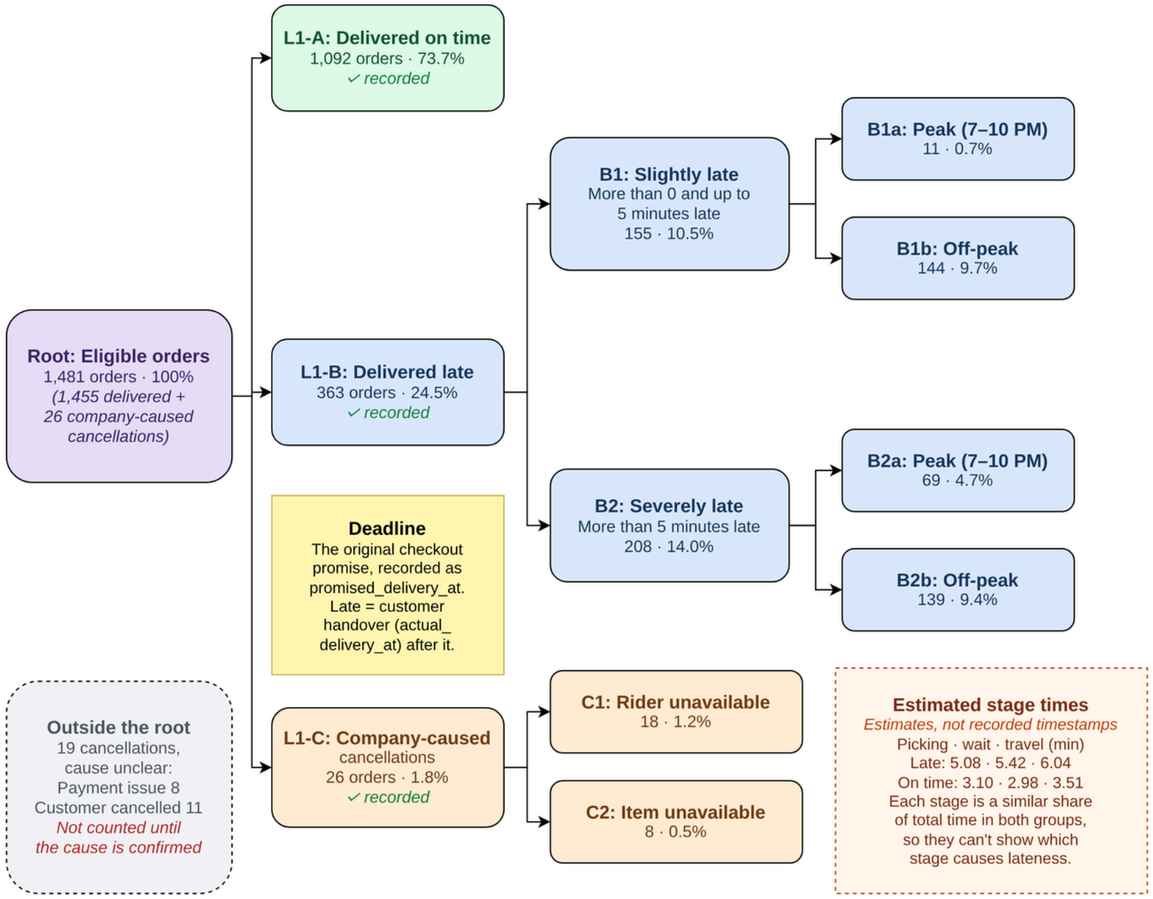
Average delivery: 11.31 min • P90 17.2 min

Provisional: 72.8–73.7%, depending on what caused the 19 unresolved cancellations.

GUARDRAILS

- Average delivery time was 11.31 minutes, and 90% of deliveries arrived within 17.2 minutes.
- Average delivery time promised: 10.33 min.
- Cancellation rate = (45 cancelled / 1,500 orders) x 100 = 3.0%

Where are orders failing to meet the delivery promise?



- **North Star (provisional)** = L1-A ÷ (L1-A + L1-B + L1-C) × 100 = 1,092 ÷ 1,481 = 73.7%. If all 19 unclear cancellations were company-caused: 72.8%. On time among delivered orders: 75.1%.
- **Each box's %** = its order count ÷ 1,481 × 100. Counts add up exactly; a child % may differ from its parent by 0.1 because of rounding.
- **Orders sheet fields used:** order_status, delay_min, is_peak_hour (peak = placed 7–10 PM) and cancellation_reason.
- **Severe** (more than 5 minutes late) is our analytical choice, not a dataset label; the peak pattern also holds at 3 and 7 minutes.
- **Scope:** 1,500 orders, 20 stores, 1–14 Sep 2026. Selection method not recorded, so results describe these orders only. Possible causes: Slide 3.

Issue tree and metric questions

Outcome → how late → peak or off-peak.The tree shows where orders miss the promise; slide 3 tests why.

HOW WE COUNT

- Source: Orders sheet fields order_status, delay_min, is_peak_hour and cancellation_reason
- Each group's % = its order count ÷ 1,481 × 100.
- On time means handed over by the original deadline; peak = orders placed 7–10 PM.

The 5-minute cut-off is our analytical choice; the peak pattern also holds at 3 and 7 minutes.

Box	How it is counted (orders, % of E)	Question it answers
B1	Late by more than 0, up to 5 min: 155 (10.5%)	How many misses are small?
B2	Late by more than 5 min: 208 (14.0%)	How many misses are large?
B1a / B1b	Slightly late, peak 11 (0.7%) / off-peak 144 (9.7%)	When do small misses happen?
B2a / B2b	Severely late, peak 69 (4.7%) / off-peak 139 (9.4%)	When do large misses happen?
C1	Cancelled, rider unavailable: 18 (1.2%)	How many orders are lost because no rider was free?
C2	Cancelled, item unavailable: 8 (0.5%)	How many orders are lost to stockouts?
Outside	Payment issue 8 and customer cancelled 11. *Recorded category; cause not confirmed*	What caused these cancellations?

Severe delays are more common at peak: 25.0% of peak deliveries were more than 5 min late (69 ÷ 276) vs 11.8% off-peak (139 ÷ 1,179).

Prioritised hypotheses and what the data shows

Ranking by size of the effect, strength of evidence and whether Blinkit can act. Results describe these 1,500 orders only.

A pattern is not proof of cause, and stage times are estimates.

25.0% of peak deliveries severely late, vs 11.8% off-peak

9.6 min peak promise, vs 10.5 off-peak, while actual time is 11.78 vs 11.20

+1.02 min estimated rider wait at peak (4.41 vs 3.39 off-peak)

Rank	Hypothesis	Test metric	Would reject it	Evidence so far	Next step
1 H1 **Supported: severe delays more common at peak**	Severe delays are more common at 7–10 PM. A store-to-rider bottleneck is one possible explanation.	Severe-delay rate (>5 min late ÷ delivered) and estimated rider wait, peak vs off-peak, within distance and rain groups.	No higher severe-delay rate at peak, or the gap disappears within groups.	Severe 25.0% vs 11.8% (+13.2 pts, 95% CI +7.8 to +18.6); holds in every distance and rain group. Late at all: 29.0% vs 24.0%, not clear (95% CI −0.9 to +10.9). Wait +1.02 min. Bottleneck possible, not established; rider shortage not shown.	Get hourly rider log-ins and recorded pickup timestamps.
2 H2 **Supported at peak; borderline in rain**	Promises don't match conditions: shorter at peak and no longer in heavy rain.	Average promise vs actual time, peak vs off-peak and by rain; late rate in heavy rain.	Promises are as long or longer where actual times are longer.	Peak promise: 9.6 vs 10.5 min off-peak; actual time: 11.78 vs 11.20. Heavy rain: promise 10.13 vs 10.36 min without rain; late rate 33.9% vs 24.8% (95% CI: 0.0 to +18.3 points).	Review how the promise is set; pilot peak and rain buffers.
3 H3 **Inconclusive; overlaps with peak**	Store-hours busier than the store's usual load run late more often.	Late rate by load quartile (orders placed that hour ÷ typical orders per hour), tested within off-peak hours.	No rise in late rate from the lowest to the highest load quartile off-peak.	Busiest vs least busy: 29.1% vs 21.4% late; off-peak: 27.1% vs 21.7% (inconclusive). All peak hours were ≥ 5% above usual demand, so peak and load effects overlap.	Test with hourly staffing and rider data.
4 H4 **No clear evidence in this analysis**	Stores with slower picking deliver late more often.	Store late rate vs average picking time (20 stores); slower vs faster half.	No clear association between picking time and late rate.	Stores with slower picking had 26.1% late deliveries vs 23.9% for faster stores. The link is uncertain (r = 0.43; both 95% CIs include zero).	Recheck with more stores or a longer period.
5 H5 **No clear evidence in this analysis**	Late deliveries drive support tickets and refunds.	Each customer's late share vs their tickets and refunds.	No clear association between late share and tickets or refunds.	No clear linear relationship between lateness and tickets (r=0.02) or refunds (r=0.08). Tickets are not tied to any specific order.	Link tickets to order IDs.
6 H6 **Inconclusive: only two new stores**	New stores are less reliable.	Late rate, stores under 180 days old vs older stores.	No meaningful gap.	New stores: 25.5% late vs 24.9% for older stores. The gap is uncertain; this does not prove equal performance.	Revisit when more new stores open.

Percentage points (pts) are the difference between two percentages. The 95% CI reflects the uncertainty around that difference. If it contains a zero, we can't be sure that the groups are different. Correlation (r) measures how two measures change together in a straight line pattern. Near zero means not a lot in a straight line.

Other checks: late trips were 32.8% of trips over 4 km and 21.6% of trips up to 1 km, but the difference is uncertain. There was little direct connection between traffic and lateness (r = 0.02). The late rate was lower on event days (14.3% versus 25.4%) but the reason is unknown.